

Neo-Cybernetics Manifesto

Intellegere, Conciliare, Evolvere

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Feedback and Contributions are welcome.

Preamble

In a world of rapid technological and societal change, how do we create systems that are resilient, adaptable, and ethically sound? As we navigate global challenges, from climate change to the implementation of advanced AI systems, society needs robust frameworks that can guide us through complexity toward greater harmony and understanding.

Neo-cybernetics offers such a framework. It equips us to understand and manage complex systems, from social and ecological networks to cutting-edge technologies, with practical tools and insights that can shape meaningful, sustainable progress.

This manifesto serves as a guide to the intentions and vision for neo-cybernetics, laying the groundwork for how we approach and engage with complexity in the 21st century.

In the true spirit of cybernetics, this document is intended to evolve through an ongoing process of adaptation. We intend to refine our approach as we gather insights from both our community and the wider world. Your feedback and collaboration are invaluable in helping us build a future that is ethical, sustainable, and accessible to all.

Definition

Neo-cybernetics is the science of communication, interaction and governance in networks and complex adaptive systems.

The term "cybernetics" originates from the Greek word 'kybernētēs', which means steersman, governor, pilot, or one who guides. Established primarily in the 1940s and 1950s by pioneers such as Norbert Wiener, cybernetics was conceived as a transdisciplinary study of systems, controls, and feedback mechanisms in natural and artificial systems.

By the 1970s and 1980s, a second order of cybernetics arose. Thinkers like Heinz von Foerster and Humberto Maturana shifted the focus towards recursion, circularity, and the role of the observer, emphasizing the reflexive, autopoietic nature of living systems.

In the decades that followed, cybernetics began to lose its interdisciplinary foothold. It was increasingly associated with science fiction or reduced to a simplistic "human-machine" paradigm, often overshadowed by technological advances that focused solely on practical applications. As a result, much of the field's holistic approach to systems thinking was set aside in favor of narrower, utility-driven perspectives.

Today, the third wave of cybernetics seeks to reclaim and expand upon these foundational ideas. Neo-cybernetics is part of this broader movement, integrating the insights of earlier waves to address the challenges of our interconnected world. It focuses on understanding how diverse systems—social, ecological, and technological—interact and evolve together. Neo-cybernetics offers a renewed perspective that goes beyond technical applications, emphasizing ethical considerations, sustainability, and the interconnectedness of human and technological progress.

The "neo-" prefix signals a renewed commitment to cybernetics as a transformative discipline that bridges traditional scientific silos, offering insights and frameworks for a more holistic and adaptable future.

Description

Neo-cybernetics is part of the third wave of cybernetics, expanding and reshaping foundational ideas to address today's complex challenges. This approach emphasizes a holistic view that incorporates non-linear behaviors, emergent

properties, causal loops, human-machine collaboration, ethical and ecological values, and future-oriented thinking.

Rather than viewing systems as mere wholes, neo-cybernetics takes a multi-dimensional approach. It examines individual components, the ecosystems they inhabit, and the extensive networks of interactions they form. This perspective helps unravel the complex flows of resources, dependencies, and feedback loops, providing deeper insights into both natural and engineered systems.

Historically, understanding complex systems was limited by the boundaries of specialized knowledge and the constraints of analytical tools. However, advances in communication systems and AI are now helping to bridge these gaps. As information becomes more accessible and interpretable across disciplines, it is easier to connect insights from fields like biology, economics, and social sciences, creating a more integrated approach to problem-solving. These tools not only make information available to a broader audience but also allow for the synthesis of diverse knowledge areas, fulfilling the interdisciplinary aspirations of neo-cybernetics.

The remarkable progress in fields like AI, quantum computing, renewable energy, biotechnology, materials science, and neuroscience demonstrates the level of advancement our society has achieved. Each breakthrough builds upon decades of technological and scientific knowledge, highlighting how interconnected progress across disciplines and generations can drive innovation. By leveraging existing technologies and shared knowledge, society has reached new heights in understanding and reshaping the systems around us.

As we approach the fifth industrial revolution, which promises to reshape economic and social structures, neo-cybernetics intends to be at the forefront. Its aim goes beyond adapting to change; it seeks to guide it. By designing models that foster harmony across diverse aspects of life, neo-cybernetics envisions a future where technology supports individual aspirations while enhancing collective well-being.

In pursuit of this goal, neo-cybernetics is dedicated to aiding governments, businesses, and communities in their journey toward Society 5.0, a vision of a future where technological potential and human values coexist in balance.

Characteristics

Neo-cybernetics is characterized by its holistic understanding of complex systems, a multi-level approach to emergent behaviors, and an integrative perspective that

spans across multiple fields. Below, is a summary of the foundational approaches and driving principles of the discipline.

Convergence of Cybernetic orders

Neo-cybernetics combines insights from first-order Cybernetics (feedback, control, mechanistic couplings), second-order (observer, recursion), and third-order cybernetics (human-machine-environment co-evolution). Tracing its roots to foundational concepts from the 1940s, it encapsulates the evolutions of thought across orders.

However, it moves beyond viewing these as strictly progressive, recognizing limitations in earlier orders.

Liberated from militaristic and behaviorist associations, neo-cybernetics integrates human, machine, and environmental systems. It is ethically attuned through emphasizing symbiosis and co-constructed environments. This holistic approach ensures comprehensive understanding of cognitive systems within broader socio-cultural contexts. With expanded boundaries, neo-cybernetics aims for a more ethically grounded and ecologically sustainable integration.

Advanced Interdisciplinary Integration

Neo-cybernetics champions the fusion of insights from modern fields, including technology ethics, biotechnology, machine learning and deep learning, quantum computing, neurosciences, ecology, futures studies, and systems theory. This synthesis aims to offer a comprehensive, multi-faceted perspective, ensuring that the discipline remains at the forefront of understanding and navigating complex adaptive systems in an ever-evolving landscape.

Complex Adaptive Systems Focus

This approach concentrates on recognizing and understanding non-linear dynamics and emergent behaviors in systems that continuously adapt and evolve. Such a perspective becomes increasingly crucial in the face of rapidly changing societal and technological landscapes, where simple cause-and-effect relationships often give way to unforeseen outcomes.

Futures Studies Integration

Placing a significant emphasis on exploring possible futures, incorporating foresight methodologies, advanced simulations, and principles of futurism. This proactive approach aids in anticipating system evolutions, facilitating strategic planning, and preparing for diverse eventualities in an increasingly uncertain, complex and interconnected world.

Ethical and Sustainability Lens

Drawing inspiration from both cybernetics and ecology, neo-cybernetics critically examines systems with a heightened focus on ecosystems and their interrelations. It emphasizes the ethical implications of systems, especially concerning technology, the environment, and societal impacts. By intertwining ecological understanding with systems thinking, it champions a holistic approach to sustainable decision-making and strategies. The goal is not just to ensure that systems function optimally but also to ensure that they do so in a manner that is harmonious with the environment, ethically sound, and beneficial for society at large.

Human-AI Collaboration

Delving deep into the synergistic relationship between humans and AI systems, this focus area examines the potential for mutual evolution, collaboration, and co-regulation.

It seeks to understand how these two distinct entities can harmoniously interact, learn from one another, and jointly shape the future. Recognizing the unique strengths and limitations of both, the aim is to craft an integrated framework where human intuition and creativity seamlessly merge with AI's computational prowess and data-driven insights.

Network Theory Integration

Probing the structure of networks, this domain seeks to understand their decentralized character, unfolding behaviors, and emergent properties.

It emphasizes the significance of nodes and links, exploring how individual components interact within the larger network structure. In doing so, it unravels patterns, cascading effects, and vulnerabilities. As our world becomes increasingly interconnected — from social networks to technological infrastructures — harnessing insights from network theory becomes paramount to predict, manage, and optimize these complex systems.

Holistic Analysis Tools

Leveraging cutting-edge analytical and computational methodologies, this focus area champions a comprehensive approach to understanding and modeling systems.

Rather than being confined by the constraints of traditional techniques, these tools embrace the complexities inherent in modern systems. By doing so, they offer a more integrative perspective, capturing the interdependencies, feedback loops, and non-linear dynamics that classical methods might overlook. In an era where

challenges are multifaceted and interconnected, these holistic tools are crucial for accurate prediction, intervention, and innovation.

Metasystem Transition Theory

Diving into the transformative journey of systems, this area of focus deciphers how systems undergo evolutionary leaps to form more complex structures.

By assimilating and organizing individual components and nested subsystems, a system can transition to a higher level of complexity, paving the way for novel functionalities and behaviors.

Recognizing these metamorphoses is pivotal in anticipating the next phases of systemic evolution and harnessing them effectively.

Emphasis on Meta-Learning

Meta-learning, or "learning how to learn," has become an important area in both AI development and understanding human cognition. By studying the process of learning itself, we can improve learning strategies, refine AI training methods, and deepen our insights into how humans and machines acquire knowledge.

Cultural and Societal Dynamics

This focus area examines how cultural values, norms, and practices shift as technology advances. By understanding these changes, we can better assess the social and cultural impacts of the digital age and anticipate how the interconnected global community might evolve.

Attention to Micro and Macro Scales

Our approach balances detailed, micro-level analysis—like cellular dynamics or individual behaviors—with broader, macro-level structures, such as global ecosystems and digital networks. As generative AI advances, it opens new possibilities for research, helping us connect insights across different scales and build a more comprehensive understanding.

Adversarial Dynamics Analysis

Including the study of adversarial behaviors across various levels, from individual entities to vast systems to craft robust and resilient systems that can withstand and adapt to adversarial challenges. Recognizing that oppositional forces can shape and influence systems, this aspect dives deep into understanding conflict dynamics, competitive strategies, and the ensuing impacts on system equilibrium and evolution.

Feedback Evolution

Feedback loops, often the heartbeat of complex systems, are not static. Our focus in this area centers on the dynamic nature of these loops, exploring how feedback mechanisms themselves transform as systems mature and environments shift. Recognizing these evolutionary patterns is not just an academic pursuit; it's crucial for anticipating system responses and optimizing their performance over time.

Resilience and Robustness

In a world of uncertainty, designing systems that can weather storms is paramount. Our emphasis here is twofold.

Firstly, we explore resilience, ensuring systems can adapt and recover from disruptions. Secondly, we prioritize robustness, aiming to build systems that remain unwavering even in the face of unexpected challenges.

These twin pillars guide our approach, ensuring that systems are not only adaptable but also enduringly stable, striking a balance between flexibility and steadfastness.

Focus on Connectivity Over Hierarchy

In the digital age, decentralized networks and systems call for a shift from traditional hierarchical views to a focus on connections. The neo-cybernetic approach emphasizes the relationships and interdependencies within these networks, providing a clearer view of their behavior and potential. This perspective aligns with the flexible, interconnected nature of modern systems, moving beyond rigid structures to embrace the complex links that shape them.

Challenges

As AI systems become more advanced, approaching the theoretical concept of General Artificial Intelligence (AGI), we stand on the edge of transformative capabilities that could significantly enhance decision-making and help address some of the world's most complex challenges, from existential risks to entrenched social and environmental issues.

However, the growing centrality of data in this new era also brings a heightened risk of concentrated control. If technological systems are dominated by a few powerful entities, global decision-making could be driven by narrow interests, leading to biases that overlook the diverse needs of communities and ecosystems worldwide.

Neo-cybernetics recognizes that technology's influence reaches across societal, cultural, economic, and ecological spheres, impacting both local communities and

global networks. By striving for a balanced, holistic understanding, it seeks to integrate the study of individual components with the dynamics of vast systems.

This proactive approach emphasizes the need for foresight and impact assessment, using historical data, scenario planning, and exploration of future possibilities to anticipate long-term effects. Neo-cybernetics advocates not just for responsive solutions but for preventive strategies that minimize risks, ensuring that opportunities for positive change are maximized.

In this way, neo-cybernetics envisions a world where potential pitfalls—such as unchecked global risks—are avoided, and emergent possibilities for human and ecological well-being are fully realized.

The Neo-Cybernetics Triad

The Neo-Cybernetics Triad—**Understand, Harmonize, Evolve**—captures the core philosophy of the discipline. Beginning with a deep understanding of systems, it emphasizes blending knowledge with ethical values and prioritizes continuous adaptation. This triad acts as a guide, helping us navigate, shape, and thrive within both natural and artificial systems amid constant change.

Intellegere, to Understand

At the heart of neo-cybernetics is a commitment to deep comprehension. To engage with complex systems, we must first understand how their parts interact, the forces that shape them, and the broader impacts they have. This principle emphasizes not just technical knowledge but a shared understanding that benefits everyone.

Neo-cybernetics encourages making knowledge accessible, fostering a culture where insights are widely shared and understood, creating value across communities and empowering individuals.

Conciliare, to Harmonize

Understanding alone isn't enough; insights need a purpose. Neo-cybernetics promotes the careful integration of knowledge, technology, societal values, and respect for the environment. This process involves actively identifying biases and unearthing tendentious opinions that may distort or limit perspectives. Harmonizing means not only bringing together diverse viewpoints but also critically examining them to ensure they reflect a balanced understanding. By doing so, this approach fosters inclusivity and unity, encouraging solutions rooted in respect for differences and a commitment to shared, unbiased progress.

Evolvere, to Evolve

The final pillar, *Evo/ve*, recognizes that staying static in a dynamic world leads to stagnation. Neo-cybernetics promotes adaptability and resilience, encouraging systems to grow and change in ways that benefit humanity. This approach doesn't just observe changes but actively seeks to guide them, steering systems towards paths that align with broader, collective goals. It's about proactive evolution—ensuring that as we adapt, we do so in ways that promote the well-being of all.

Key Concepts

Presented below are pivotal terms that are central to the discourse and exploration of neo-cybernetics as a discipline.

Adaptation

The capability of a system to adjust its internal structure or behavior to fit environmental challenges.

Agent

An autonomous entity that can make decisions based on their own perceptions, and act upon its environment to achieve specific goals or purposes. Agents can be individuals, groups, natural or synthetic entities, and their actions can be influenced by both internal components and external conditions.

Autopoiesis

A term primarily related to living systems, indicating the self-producing and self-maintaining nature of such systems.

Boundaries

Every system has boundaries that separate it from its environment. However, these boundaries can be permeable, allowing exchanges.

Circular Causality

This concept diverges from linear causality, where one event directly leads to another. In circular causality, feedback loops exist such that cause and effect may be interchanged, leading to patterns where an effect can also influence its cause, creating cyclical patterns of behavior.

Co-Evolution

The simultaneous evolution of two or more entities that influence each other's development.

Complexity

The attribute of systems marked by relationships, dependencies, and interactions between their components. Such systems often exhibit emergent properties and behaviors, which are challenging to anticipate or model based solely on an understanding of individual elements.

Communication

The act of transmitting, conveying, sharing or exchanging data, information, ideas, or feelings between entities.

Consensus

An agreement or unified understanding reached by a group, typically after discussions or negotiations, where members align on a particular decision or perspective.

Cybernetics

Originating from the Greek word "kubernētēs," meaning "steersman" or "governor," Cybernetics is the interdisciplinary study of systems' structures, constraints, and possibilities. It explores the mechanisms of control, regulation, and communication within and between diverse systems, whether they are biological (like animals and human bodies), technological (like machines and computer networks), or socio-cultural (like organizations and communities). By analyzing feedback loops and emergent behaviors, Cybernetics seeks to understand how systems maintain stability, adapt to changes, and evolve over time. The discipline bridges the gap between the animate and inanimate, providing a unified framework for understanding the fundamental principles that govern dynamic systems in various domains.

Data

Unprocessed and uninterpreted facts, figures, symbols, or observations that can manifest in various forms, such as numerical, textual, audio, visual, or sensory. Lacking context or meaning on its own, data serves as the foundational layer upon which information and knowledge are built through analysis and interpretation.

Emergence

A phenomenon wherein the combined actions and interactions of individual components within a system give rise to collective behaviors or properties that cannot be easily deduced from studying each part in isolation. Emergent behavior often appears unexpected and can be more sophisticated than what the individual components suggest.

Energy

The capacity to do work or cause physical change; can exist in potential, kinetic, thermal, electrical, chemical, nuclear, or other forms.

Entropy

In thermodynamics, it relates to the degree of disorder or randomness in a system. In information theory, it deals with the uncertainty of information.

Equifinality

The principle that in open systems, a given end state can be reached from different initial conditions and in various ways.

Eigenbehaviors

This term denotes the characteristic patterns of behavior that a system will naturally gravitate towards over time. Eigenbehaviors are stable states or recurring patterns for a system.

Feedback Loops

Cyclic processes within systems that facilitate self-regulation by monitoring and adjusting based on specific outputs or results. Feedback can either be reinforcing, which amplifies deviations or changes, or balancing, which works to counteract and stabilize those deviations.

Flow

The movement or transfer of resources, information, energy, or matter from one part of a system to another or between two or more systems. Flows facilitate interaction, transformation, and exchange within and between systems and can be cyclical or linear, depending on the context.

Hierarchy

A structured arrangement of systems where entities can be part of larger wholes while also containing smaller components within. Each hierarchical level has its distinct set of characteristics and functions, operating at varying scales. The interactions between these levels often influence the behavior and properties of the overarching system.

Heterarchy

A system of organization where elements are unranked or possess the potential to be ranked in multiple ways.

Holacracy

A method of decentralized management and organizational governance, where decision-making is distributed throughout self-organizing teams rather than a top-down hierarchy.

Holonomy

A concept stemming from both philosophy and systems theory, holonomy underscores the importance of viewing a system or entity holistically. Rather than perceiving the whole as a mere aggregation of its parts, holonomy emphasizes that the whole's properties arise not only from the parts themselves but crucially from the interactions and relationships among these parts. The essence of holonomy is the belief that the entirety of a system is greater than the sum of its individual components.

Homeostasis

A self-regulating process by which a system actively adjusts its internal conditions to maintain stability or equilibrium, even when faced with external disturbances. This adaptive mechanism ensures consistency in the system's functions and performance over time, resisting disruptive changes from the environment.

Hypercycles

Hypercycles are a model to elucidate the self-organization and evolution of primitive life forms. Comprising various self-replicating units (like RNA/DNA strands) interconnected in a cyclical, autocatalytic network, each replicator aids in the replication of its successor. This self-sustaining cycle is more efficient than individual replication, allowing replicators to surpass the Eigen error threshold. Thus, hypercycles are a conjectured step in the molecular evolution leading to life.

Industry 5.0

The fifth phase in the evolution of industrial development, following the fourth industrial revolution (or Industry 4.0). Industry 5.0 emphasizes the integration of human creativity and craftsmanship with advanced robotics and AI technologies. Unlike Industry 4.0, which largely focused on full automation, Industry 5.0 seeks to achieve a harmonious collaboration between humans and machines. This progressive era aims to bridge the gap between large-scale mass production and bespoke individual customization, leveraging advanced technologies such as Big Data, artificial intelligence, and virtual reality to enable a society that is human-centric, sustainable and resilient.

Information

Derived from raw data, information is processed and organized in a manner that adds meaning, relevance, and context. It serves as the foundation for acquiring

knowledge, enabling informed decision-making, and providing deeper insights into various phenomena.

Matter

Physical substance which occupies space and possesses mass, distinct from energy.

Negentropy

In thermodynamics, it refers to the degree of order or organization in a system. In information theory, it describes the measure of orderliness or predictability of a set of data.

Network

A system of interconnected nodes and links, often studied using graph theory, a branch of mathematics that explores relationships between connected entities. In the context of neo-cybernetics, a network could refer to technological infrastructures, social connections, neural pathways, or any other system where entities (nodes) are connected in various ways (links). Networks emphasize the importance of relationships and the flow of information, resources, or influence. Modern interpretations and analyses of networks can leverage graph databases, which are specifically designed to handle complex network structures efficiently.

Non-linearity

A characteristic of systems where the relationship between inputs and outputs isn't strictly proportional, often resulting in outcomes that aren't immediately intuitive. Non-linear systems can exhibit dramatic changes from small perturbations or display resilience against large disturbances, making their behavior challenging to predict solely from initial conditions.

Observer

Central to systems theory and second order Cybernetics, the observer is the entity that discerns, decodes, and engages with a system. The perspective and inherent biases of the observer can shape the interpretation and understanding of the system, highlighting the significance of subjectivity in system analysis and response. This concept underscores the idea that observations are not purely objective; they are influenced by the observer's context, knowledge, and predispositions.

Society 5.0

Society 5.0 represents the vision for the next stage of human societal evolution following the Information Society (Society 4.0). In Society 5.0, advanced technologies, particularly from the realms of AI, big data, IoT, and robotics, are seamlessly integrated into every facet of human life. The primary aim is to create a human-centric society that balances economic advancement with the resolution of

pressing societal issues such as aging populations, urbanization, and environmental challenges. Through the fusion of the cyber and physical spaces, Society 5.0 endeavors to enhance the quality of life for all, promoting a more inclusive, sustainable, and resilient future.

Swarm Intelligence

A collective behavior exhibited by decentralized, self-organized systems, typically consisting of a population of simple agents interacting with one another and their environment. Such systems can achieve complex tasks without centralized control or explicit instructions, drawing from the interconnected knowledge and adaptability of individual agents.

Teleology

Teleology, derived from the Greek words "telos" meaning "end" or "purpose" and "logos" meaning "study of," refers to the philosophical study of purpose, direction, or goals in nature. It posits that entities, behaviors, or processes exist or occur due to a predetermined objective or purpose. In the context of neo-cybernetics, teleology might be invoked to explore the inherent or designed purposes of complex systems, questioning if and how technological and biological systems pursue goals, adapt to achieve certain ends, or can be directed towards beneficial outcomes for society at large. This contrasts with purely mechanistic or deterministic viewpoints, emphasizing instead the role of objectives and intent in shaping system behaviors and trajectories.

Wicked Problems

In the context of social policy planning, "wicked problems" describe complex challenges that defy straightforward solutions due to their interdependencies, multifaceted nature, and the evolving criteria for success. These problems often resist definition, and solutions to them can lead to further complications. Unlike "tame" or "benign" problems, wicked problems lack clear stopping rules, meaning there's no definite solution. Attempts to address a wicked problem can change the understanding of the problem itself. Examples include climate change, poverty, and geopolitical conflicts. Addressing wicked problems usually requires interdisciplinary approaches, iterative processes, and adaptive strategies.

Principles

At the heart of neo-cybernetics lies a set of guiding beliefs that shape its approach and methodology. These principles form the moral and philosophical bedrock of the discipline, ensuring that it remains true to its foundational ideals even as it evolves and adapts to an ever-changing world.

Integrity

To uphold the highest standards of honesty, transparency, and ethical behavior in all endeavors.

Inclusivity

To embrace diverse viewpoints and nurture global collaboration for richer, deeper insights.

Education

To disseminate knowledge and ensure that the principles of neo-cybernetics are accessible to all.

Collaboration

To work hand-in-hand with other disciplines, understanding that unity yields richer insights.

Innovation

To continuously seek novel solutions and push the boundaries of what's possible.

Responsibility

To act as stewards of the planet, ensuring that technological advancements are in harmony with the global and local needs of society and with nature.

Commitments

While principles provide the foundational beliefs of neo-cybernetics, commitments represent the tangible promises our community makes to the world. They are actionable pledges, ensuring that the tenets of neo-cybernetics translate into real-world impact and positive change.

Universal Tech Access

Ensure that technology, in all its myriad forms, is accessible universally, transcending barriers of geography, economics, or culture.

Information Equity

Commit to making high-quality, accurate information universally available, democratizing knowledge and empowering communities everywhere.

Global-Local Nexus

Recognize and honor both the narratives of global movements and the nuances of local realities, striking a harmonious balance in our endeavors.

Cultural Confluence

Champion the confluence of diverse cultures, believing that embracing varied perspectives fosters richer insights and more holistic solutions.

Interdisciplinary Unity

Actively converge multiple disciplines, recognizing that the most profound insights and solutions often emerge from interdisciplinary collaboration.

Unyielding Resilience

Architect systems that stand resilient in the face of adversities, possessing the robustness to recover and thrive amidst challenges and failures.

Ethical Advocacy

Spread and champion awareness of the ethical dimensions inherent in our actions. Strive, always, for solutions that are both ethically sound and ecologically sustainable.

Human-Centered Synthetics

Design synthetic systems and consumables with a human-centric approach, ensuring that they enhance and respect human well-being and experience.

Join the Neo-Cybernetic Community

In an age where the complexities of our world challenge traditional paradigms, neo-cybernetics provides a guiding light.

Our discipline thrives through shared insights and collective exploration.

Engage with our ideas, contribute to the discourse, and help shape a future rooted in interconnectedness and mutual understanding.

Embrace the principles of neo-cybernetics: cultivating comprehension, harmonizing diverse visions, and evolving with purpose.

Through understanding, unity, and progress, we propel humanity forward.

Literature

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